

Summary

This study mainly investigates two questions.

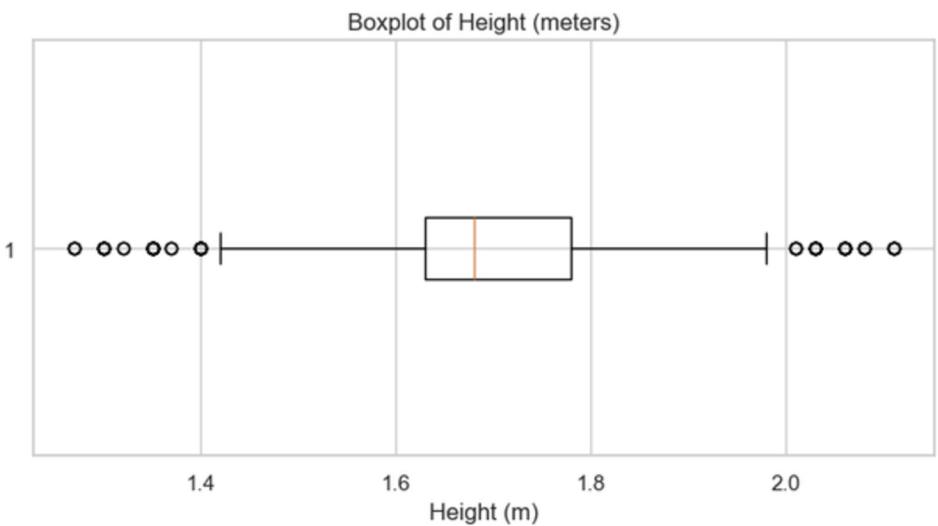
First, is the proportion of high school students who feel sad or hopeless significantly different from 0.30?

Second, is the average height of high school students different from 1.70 meters?

For the first question, the study first recoded students who felt sad or hopeless as 1 and those who did not as 0, while removing invalid values to ensure that the analysis would not be affected by erroneous data. From the cleaned data and graphs, the proportion of students in the sample who felt sad or hopeless was close to 0.30. In addition, the statistical test produced a p-value of 0.9380, indicating that the null hypothesis could not be rejected. Therefore, it can be inferred that the proportion of high school students who feel sad or hopeless is not significantly different from 0.30.

For the second question, the study used the mean, median, boxplot, and Q-Q plot to examine the distribution of height data among high school students. The results showed that the distribution was approximately symmetric. Although a small number of outliers were present, they did not affect the overall representativeness of the data. Moreover, most points in the Q-Q plot were close to the diagonal line, suggesting that the data were approximately normally distributed, which supports the use of a one-sample t-test. The test results showed that the average height of high school students was about 1.6940 meters, with a p-value of 1.941e-11, indicating statistical significance. Therefore, the null hypothesis could be rejected, meaning that the average height of high school students is significantly different from 1.70 meters.

Overall, this study found that the proportion of high school students who feel sad or hopeless is not significantly different from 30%, while the average height is significantly different from 1.70 meters.



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